Plato's Five in the Octet

All five Platonic solids nested in every cell, and a golden ladder of crystals to the quasilattice



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DICTO proposed putting all five Platonic solids into the octet with pieces of the solids themselves between them, nesting them in several combinations and at several scales, the supercells from the 120-cell, the 600-cell and the middle slice (his UNITY), filling the gaps with harmonics, waves and golden skin, and the ladder to a universal lattice. Computation, proofs and drafting with Claude (Anthropic); an independent review (Claude) checked the prior art and supplied the proofs of Theorem 3.

Plato's five solids do not fill space together. But the octet, the honeycomb of tetrahedra and octahedra, can hold all five at once: inside every octahedron sits an icosahedron, inside that a dodecahedron, then a cube, then a tetrahedron, and every tetrahedron of the octet holds the same nest at half the size. The nest goes on for ever, a third smaller at each turn. And a ladder of crystals, each a golden ratio larger than the last, climbs from the octet's cubic order to the five-fold order of quasicrystals, with a nest on every icosahedral seat.

Abstract. Theorem 1 (Five in the octet). In the octet of edge 1 every octahedron holds the nest octahedron > icosahedron (Pacioli, edge $\sqrt{2}/\varphi^2$) > dodecahedron (edge $\sqrt{2}/(3\varphi)$) > cube (edge $\sqrt{2}/3$) > tetrahedron (edge 2/3); the nest's cube is exactly the octahedron's dual cube, so the dodecahedron touches the octahedron at its eight face centres. Every tetrahedron of the octet holds the same nest at half size through its midpoint octahedron, with the truncated tetrahedron (edge 1/3) between. All five Platonic solids sit in every cell of a structure periodic in three directions. **Theorem 2 (For ever).** The nest repeats inward and outward by exactly 1/3 per turn, keeping its orientation: the golden factors cancel; the shorter loop cube > icosahedron > dodecahedron > cube repeats by $\varphi/3$. Exact in $\mathbb{Q}(\sqrt{2}, \sqrt{5})$. We list the twelve classical inscriptions, the fourteen chains through all five (eighteen with the truncated tetrahedron), the gap pieces (two of them new shapes) and supercells from the 120-cell, the 600-cell and the middle slice; the Dodeca-13 fits inside its own central dodecahedron at exactly 1/3. **Part 2.** On the cubic approximants of the icosahedral quasilattice (Elser and Henley 1985), rungs 1/1 to 13/8, every cell is tiled by golden rhombohedra, and **Theorem 3** gives points = tiles per cell by an exact count (16, 136, 576, 1220, 10 336, 43 784); the icosahedral seats are the triacontahedral vertices, at least $\sqrt{3}\,\varphi^2$ apart, and each carries a nest. Diffraction measures the climb: ten-fold score 0.916 at 1/1, 0.9976 at 13/8, 0.9977 for the quasilattice.

1. Five in the octet

Theorem 1 (Five in the octet). Let the octet have edge 1. (a) Every octahedron O holds regular solids O ⊃ I ⊃ D ⊃ C ⊃ T: I with its 12 corners at the golden points of O's 12 edges (Pacioli), edge $\sqrt{2}/\varphi^2$; D on the 20 face centres of I, edge $\sqrt{2}/(3\varphi)$; C on 8 corners of D, edge $\sqrt{2}/3$; T on 4 alternate corners of C, edge 2/3. (b) C is O's dual cube: its corners are the 8

face centres of O. (c) Every tetrahedron of the octet holds the octahedron of its edge midpoints (edge 1/2), and with it the nest of (a) at half size; between them sits the truncated tetrahedron with corners at the thirds of the edges (edge 1/3). (d) C − T is four copies of O's centre pyramid scaled by 2/3. Hence every cell of the octet holds all five Platonic solids, and the whole is periodic in three directions.

Proof. Place O with corners (±1/√2, 0, 0) and cyclic. The points (0, ±a, ±b) and cyclic with b = φa, a + b = 1/√2 lie on O's edges in the golden ratio and are the corners of a regular icosahedron of edge 2a = √2/φ² (Pacioli). Each face of O contains three of these points, so eight faces of I lie in O's face planes: I and O share their inscribed sphere and their eight face centres on the three-fold axes. The 20 face centres of I are the corners of the dual dodecahedron D; eight of them are those shared face centres, (±t, ±t, ±t) with t = 1/(3√2), which are the corners of a cube (Euclid XIII.17: a cube on eight corners of a dodecahedron) and are O's face centres, proving (b); its edge is 2t = √2/3, and the alternate corners give T with edge 2t√2 = 2/3. For (c), the midpoints of a regular tetrahedron's edges form an octahedron of half the edge, and cutting the corners at one third gives the Archimedean truncated tetrahedron, which contains the midpoints. For (d), a corner of C outside T and its three neighbours in T form a trirectangular tetrahedron with legs √2/3, and O's centre pyramid is trirectangular with legs 1/√2. The octet is periodic (fcc), so the nests are too. Every step was checked in exact arithmetic in Q(√2, √5) (proof/exact_nest.py; all factors and sizes of §§2–3 in proof/exact_audit.py). ∎

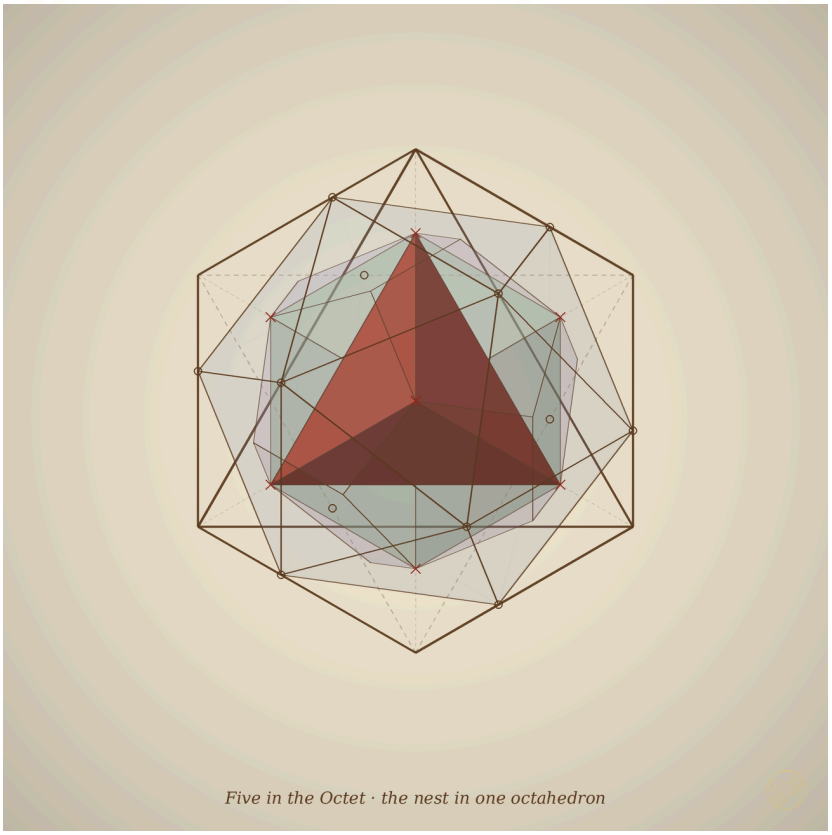


Figure 1. The nest in one octahedron, seen down a three-fold axis: the octahedron in sepia line, the icosahedron, dodecahedron and cube as tinted glass, the tetrahedron solid. Rings: the golden points of the octahedron's edges, where the icosahedron's corners sit (Pacioli); crosses: the octahedron's face centres, which are the cube's corners (Theorem 1b).

Theorem 2 (For ever). (a) The chain O ⊃ I ⊃ D ⊃ C ⊃ T ⊃ O′, with O′ the midpoint octahedron of T, has edge factors √2/φ², φ/3, φ, √2 and 1/2, whose product is exactly 1/3; O′

> has O's orientation. So the nest repeats inward, and by reversal outward, a third at each turn. (b) The loop C ⊃ I ⊃ D ⊃ C′ (I touching all six faces of C, edge 1/φ; D on I's face centres; C′ on eight corners of D) has factors 1/φ, φ/3, φ, product φ/3, and C′ has C's orientation: a golden nest for ever.

Proof. Multiply the factors of Theorem 1 and of the inscriptions in §2; the golden numbers cancel in (a). The orientations follow from the corners: O′ has its corners on O's axes, C′ has corners (±t, ±t, ±t) like C. Exact check proof/exact_loops.py. ∎

2. Combinations and gap pieces

Twelve classical inscriptions join the five solids (corners on corners, on edge points or on face centres), with these edge factors: O ⊃ I $\sqrt{2}/\varphi^2$ (Pacioli); O ⊃ C $\sqrt{2}/3$ (face centres); O ⊃ T 2/3 (four face centres); I ⊃ D $\varphi/3$ (face centres); I ⊃ O $\varphi/\sqrt{2}$ (six edge midpoints on the axes); D ⊃ C φ (eight corners); D ⊃ I $\varphi^2/\sqrt{5}$ (face centres); D ⊃ T $\varphi\sqrt{2}$ (four corners); C ⊃ I 1/φ (touching the faces); C ⊃ O $1/\sqrt{2}$ (face centres); C ⊃ T $\sqrt{2}$ (four corners); T ⊃ O 1/2 (edge midpoints). With the truncated tetrahedron U: T ⊃ U 1/3, U ⊃ O 3/2, U ⊃ T 5/3. These give fourteen chains through all five solids, and eighteen through all six with U (data-nest-catalogue*.txt). Both cells of the octet carry a complete chain: T ⊃ O (1/2) continues as O's chain at half size.

gap	pieces
O – I	the six Admiral Caps (DICTO, F21), one round each corner of O
I – D	12 Sheriffs (DICTO's name: the pentagonal cap, a corner of I over each face of D; base edge = D's edge, slant edges exactly $\sqrt{3}/\varphi$ × base) and 30 Lone-stars (DICTO's name: the thin wedge, an edge of I with the edge of D beneath it, crossing at right angles; its other four edges equal). New shapes.
D – C	Euclid's six roofs
C – T	four copies of O's centre pyramid at 2/3 (Theorem 1d)

3. Supercells

Route of the 4D structures: the 120-cell's first shell, 1 + 12 dodecahedra, is DICTO's Dodeca-13 (13 dodecahedra, 30 wedges, 20 needles); each dodecahedron holds the chain D ⊃ I ⊃ O ⊃ C ⊃ T (edges 1, $\varphi^2/\sqrt{5}$, $\varphi^3/\sqrt{10}$, $\varphi^3/(3\sqrt{5})$, $\sqrt{2}\,\varphi^3/(3\sqrt{5})$), and with each outer nest the mirror image of the central one in their common face, all 13 nests are regular and nested; six outer octahedra are parallel to the central one, six turned by 26.57° (numerical). The 600-cell's twenty tetrahedra round one corner, built with regular tetrahedra, are pairwise disjoint and leave 14.1% of their hull open (numerical) (the three-dimensional trace of the four-dimensional curvature); each holds T ⊃ U ⊃ O ⊃ I ⊃ D ⊃ C. The middle slice, the icosidodecahedron, with twelve J11 on its pentagons is DICTO's UNITY; each J11 with its pentagonal cap is a regular icosahedron holding a nest, the caps are disjoint (total volume $(5 + \sqrt{5})/2$), and their tips form a regular icosahedron of edge exactly $2/\sqrt{5}$ at UNITY's heart, clear of every cap.

> **Proposition.** The Dodeca-13 fits inside its own central dodecahedron at exactly 1/3 of its size, touching all twelve faces at once.

Proof. Let r be the central dodecahedron's face distance and n_k its face normals. The outer dodecahedron mirrored in face k has centre $2r\,n_k$ and its far face in the plane $p \cdot n_k = 3r$, so the cluster reaches 3r along n_k; any other outer dodecahedron j reaches at most $2r\,(n_j \cdot n_k) + R \le 2r/\sqrt{5} + R = 2.397 < 3r = 3.341$ (edge 1; R the circumradius), and the wedges and needles lie in the hull of the thirteen. So the cluster's extent in every face direction is exactly 3r, and the largest copy inside the slab $|p \cdot n_k| \le r$ is at scale 1/3. ∎

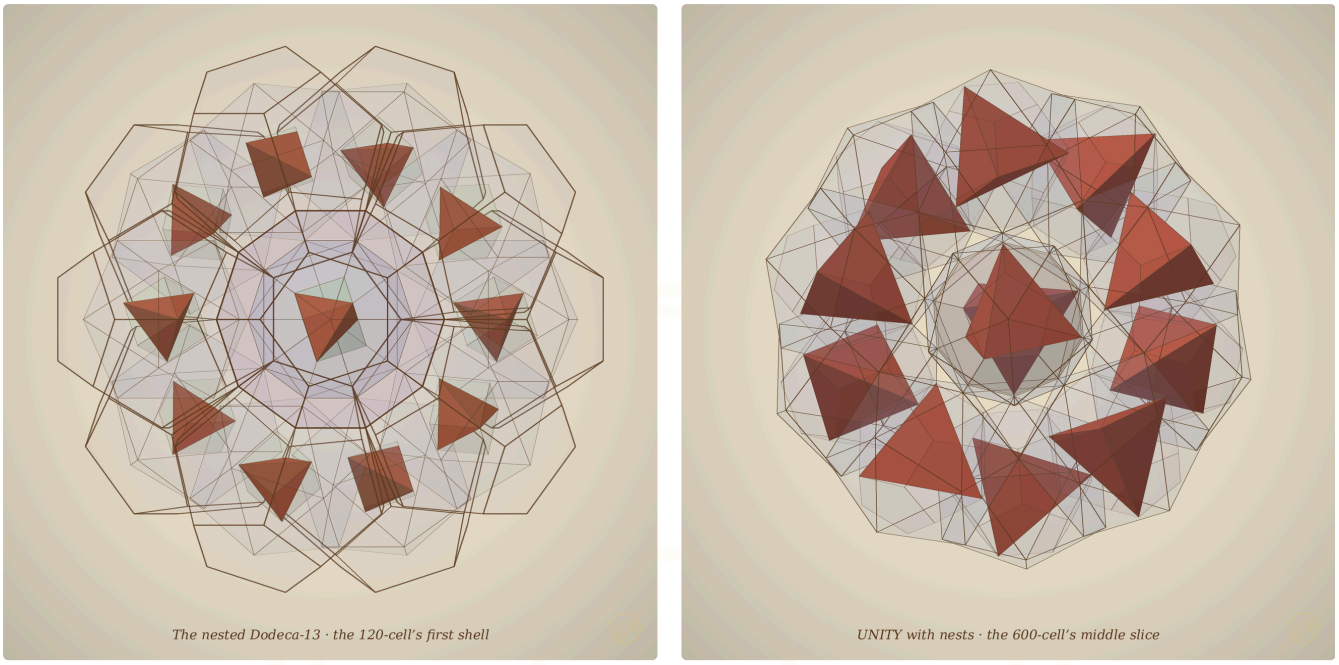


Figure 2. Left: the nested Dodeca-13 (the 120-cell's 1 + 12) down a five-fold axis, turns aligned by mirroring; the tetrahedra (solid) show each cell's nest. Right: UNITY (the 600-cell's middle slice) with a nest in each of its twelve icosahedra and the heart icosahedron (gold) formed by the caps' tips.

4. Part 2. A golden ladder of crystals

The icosahedral quasilattice is the shadow of the six-dimensional cubic lattice cut by a window, the rhombic triacontahedron, in the hidden directions (Kramer and Neri 1984); its tiles are the two golden rhombohedra. Replacing the golden number by the Fibonacci ratios 1, 1/2, 2/3, 3/5, … in the hidden directions only (Elser and Henley 1985) makes the shadow periodic: the cubic approximants, each a rung of a ladder whose cell side grows by φ.

rung	cell	golden rhombohedra per cell (prolate + oblate) = points	seats per cell	ten-fold score
1/1	body-centred cubic, side $2\varphi^2$	10 + 6 = 16	0	0.9158
2/1	cubic, side $2\varphi^3$	84 + 52 = 136	1	0.9736
3/2	cubic, side $2\varphi^4$	356 + 220 = 576	6	0.9937
5/3	body-centred cubic, side $2\varphi^5$	754 + 466 = 1220	16	0.9969
8/5	cubic, side $2\varphi^6$	6388 + 3948 = 10 336		0.9974
13/8	side $2\varphi^7$	27 060 + 16 724 = 43 784		0.9976
quasilattice	not periodic	ratio φ	share τ^{-9}	0.9977

> **Theorem 3.** On every rung: (a) points per cell = tiles per cell = $\Sigma\ |\det(g_l, g_m, g_n)| / \det \Lambda_\perp$, summed over the 20 triples of hidden directions; (b) a point whose twelve neighbours form a perfect icosahedral star with nothing else within distance 2 (Q4 = 0, Q6 = 0.6633) is exactly a vertex surrounded by 20 prolate rhombohedra, a rhombic triacontahedron; (c) two such seats are never closer than $\sqrt{3}\ \varphi^2 = 4.5346$, the prolate rhombohedron's long diagonal, and that distance occurs from 3/2 on.

Proof (outline). (a) $\Lambda_\perp$ is a lattice, the points per cell are the lattice points in the window, a tile of type ijk exists exactly when its corner's hidden image lies in the complementary parallelepiped, and the window is dissected into translates of the same twenty parallelepipeds (Shephard 1974), each holding its index of points. (b) A rhombohedron meeting the point at an obtuse corner would put a further point at distance 2; so all surrounding tiles are prolate with the point at their acute corner, twenty of them. (c) Two seats differ by a lattice vector whose hidden image lies in twice the seat window; a finite enumeration on every rung finds none shorter than $\sqrt{3}\ \varphi^2$. The seat counts 1, 6, 16 are generic values (expected 1.38, 6.8, 15.8), and the seat share tends to $\tau^{-9} = 0.01316$, the known frequency of the triacontahedral vertex (Henley 1986; Baake and Grimm 2012); on 5/3 it is already 0.0131. ∎

The tile counts in the table are exact (integer arithmetic from Theorem 3a, script exact_counts.py); the seat counts are those of our window shift, and the ten-fold scores are numerical. Each seat carries a nest: its twelve neighbours form a regular icosahedron of edge 2 holding D, C, T and O, and the twenty prolate rhombohedra round it are DICTO's golden skin; neighbouring nests never meet. On the first rung's gaps the golden skin also wins as a filling: grown in φ-steps along the icosahedra's five-fold axes it fills 82% of the gap round icosahedra on a body-centred lattice, against 54% for Platonic gravel (voxel estimates, grid 48^3).

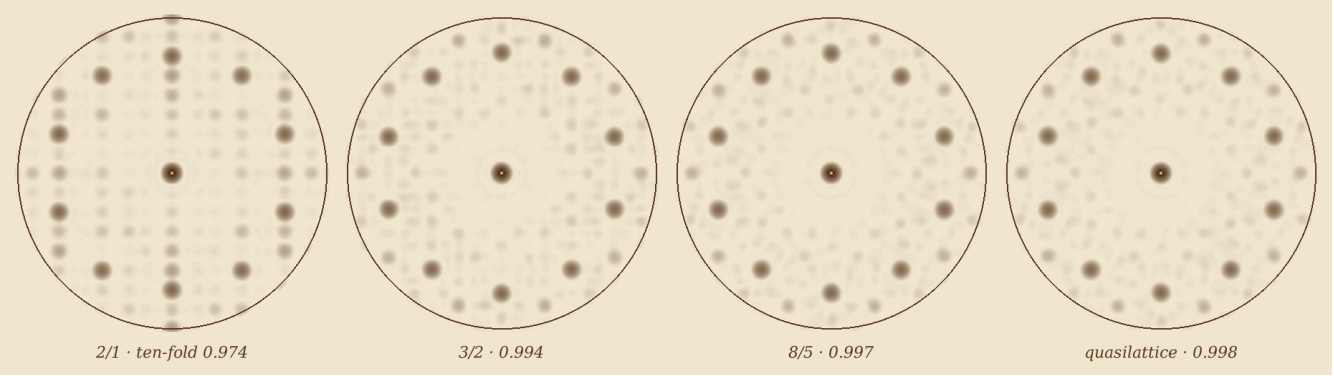


Figure 3. The climb seen in diffraction, square-on to a five-fold axis: rungs 2/1, 3/2, 8/5 and the quasilattice. Rows and columns give way to ten-fold rings; ten-fold score (the pattern's correlation with itself turned by 36°) 0.974, 0.994, 0.997, 0.998.

5. What is known and what is new

Known: Pacioli's icosahedron in the octahedron; the cube on a dodecahedron (Euclid XIII.17) and the classical inscriptions of §2; the octet; the rational approximants and their cell sides (Elser and Henley 1985; the side rule as quoted by Ishimasa 2019); the triacontahedral vertex and its frequency τ^{-9} (Henley 1986; Baake and Grimm 2012). New within the sources checked: that the nest's cube is the octahedron's dual cube, and the octet as a lattice carrying all five solids in every cell (Theorem 1); the nest for ever by thirds and the golden loop (Theorem 2); the gap pieces of I − D (Sheriff and Lone-star); the nested supercells and the

Dodeca-13 at 1/3; the exact count (a) and the seat characterisation and spacing (b, c) of Theorem 3; nests on the ladder and the measured climb. Sources checked: the reference library's full text (251 PDFs, keyword sweep; 25 read in context, including Baake–Grimm 2012, Baake 1999, Dietl–Eschenburg 2016, Hann–Socolar–Steinhardt 2016, Frettlöh 2011, Kramer 2011, Senechal 2004, Moody 2000, Kleman 2013, Ishimasa 2019, Deguchi et al. 2015, Fujita et al. 2013, Pacioli 1509, Kepler 1596 and 1619, Heath's Euclid); web: Wikipedia, Regular icosahedron, Divina proportione, Compound of dodecahedron and icosahedron, Dual polyhedron, Tetrahedral-octahedral honeycomb; the Physical Review Letters abstract of Elser and Henley 1985. Not seen in full: Elser–Henley 1985, Henley 1986, Goldman–Kelton 1993, Elser–Sloane 1987, Katz–Duneau 1986, Kramer–Neri 1984.

6. What arose unbidden

As with the Ogdoecontahedron, several things were not sought but appeared on the way: the dual cube inside the nest; in D ⊃ C ⊃ I the icosahedron has the dodecahedron's own edge; in the chain U ⊃ O ⊃ I ⊃ D ⊃ C ⊃ T the innermost tetrahedron has the outer truncated tetrahedron's edge; the third that appears twice, in the nest's turn and in the Dodeca-13 inside itself; UNITY's heart icosahedron of edge $2/\sqrt{5}$; and the two new pieces of I − D, the Sheriff and the Lone-star.

7. Open

The body-centred and simple cubic alternation of the rungs as a theorem; nests joined across seats; a comparison of the seats with the cluster centres of real 1/1 and 2/1 approximant crystals; whether this ladder closes DICTO's book The Golden Ladder, whose top rung, E8, casts exactly this quasilattice into three dimensions (Elser and Sloane 1987).

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